STA201: Elements of Statistics and Probability

Lecture 5 & 6

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Measures of Dispersion

Dispersion: Concepts

Team 1: 72 73 76 76 78

Team 2: 67 72 76 76 84

- The measures of the central tendency of the two data sets:

	Team 1	Team 2
Mean	75	75
Median	76	76
Mode	76	76

Dispersion

- We can see that both teams have the same mean, median and mode but the two teams are definitely not same.
- The characteristic that make the two data sets different seems to be the spread of the two sets of data.
- This spread of the data is known as the dispersion of the data.
- The distance of the different values from the central value is called dispersion.
- The measures of dispersion are of very much important in statistical studies.

Different Measures of Dispersion

Important and most commonly used measures of dispersion -

- Range
- Inter-quartile Range
- Mean Deviation
- Variance
- Standard Deviation

Deviation

- Coefficient of Mean Deviation
- Coefficient of Variation

- Coefficient of Range
- Coefficient of Quartile

Which measures of Dispersion to choose?

When dealing with data, if ones' objective is "only to determine" the variation of single set of variable/data s/he can choose to use absolute measure of dispersion

When dealing with data, if ones' objective is "to determine and compare" the variations of multiple set of variables/data having expressed in same/different unit(s) s/he can choose to use relative measure of dispersion

Absolute measures of dispersion

Range

- The range is the difference between the largest and smallest numbers in the dataset.
- As a primary and quick idea of dispersion, range is of great use.
  - Statistically, range, by taking into consideration only the two observations (minimum and maximum) is rated as a crude measure of dispersion.
- The measure can be influenced very badly by only one extreme observation.

  - For a set of observations $x_1, x_2, \ldots, x_n$, the following is the formula for the Range,
    Range = Highest value (H) - Lowest value (L)

The Inter-quartile Range

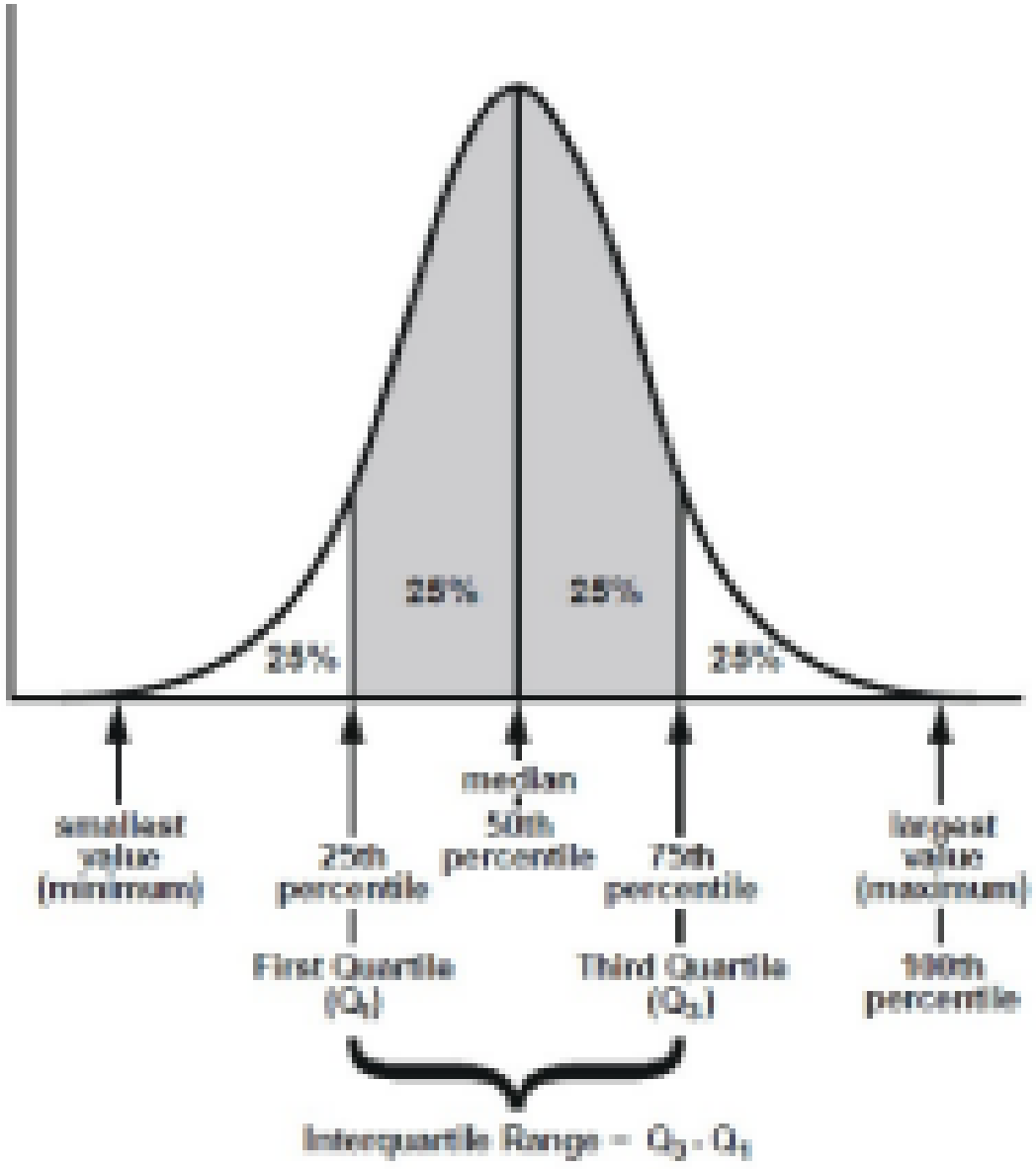
- The Inter-quartile Range is the difference between the third and the first quartile of the data set.
- In other words, the Inter-quartile range measures the spread in the central 50% of the ordered data.
- Inter quartile range, by avoiding the 25% edge observations from both side of the data becomes free from the influence of extreme observations.

  - For a set of observations $x_1, x_2, \ldots, x_n$, the following is the formula for the Inter-quartile Range (IQR),

$$\text{IQR} = Q_3 - Q_1,$$

where Q_3 = Third Quartile, Q_1 = First Quartile

Inter-quartile Range


25%
25%
25%
25%
smallest value (minimum)
25th percentile
First Quartile (Q_1)
median
50th percentile
75th percentile
Third Quartile (Q_3)
largest value (maximum)
100th percentile
Interquartile Range = $Q_3 - Q_1$

The Mean Deviation

• Mean deviation (MD) is the arithmetic mean of the absolute deviations of each of values from the mean of the data. • MD, unlike range and IQR includes all the observations in its computation.

• For a set of observations $x_1, x_2, \ldots, x_n$, the formula for the Mean Deviation,

$$MD = \frac{1}{n} \sum_{i=1}^{n} |x_i - \bar{x}|.$$

The Variance

- Variance has almost all the desirable properties of a good measure of dispersion.
- It involves all the observations, It has nice mathematical expression and can easily be used in algebraic manipulation. • Because of squaring the deviation, the units of measurement of the variance differ from units of measurement of original observation.

- For a set of observations $x_1, x_2, \ldots, x_n$, the following is the formula for the Variance,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$= \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right)$$

The variance: Formula

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$= \frac{1}{n-1} \sum_{i=1}^{n} (x_i^2 - 2x_i\bar{x} + \bar{x}^2)$$

$$= \frac{1}{n-1} \left(\sum_{i=1}^{n} x_i^2 - 2\sum_{i=1}^{n} x_i\bar{x} + n\bar{x}^2 \right)$$

$$= \frac{1}{n-1} \left(\sum_{i=1}^{n} x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \right) \quad [\because \sum_{i=1}^{n} x_i = n\bar{x}]$$

$$= \frac{1}{n-1} \left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2 \right)$$

$$\frac{i=1}{n} x_i = \bar{x}]$$

The Standard Deviation

- The sample Standard Deviation (SD), denoted by S, is the square root of the sample variance.
- Sample standard deviation includes all the observations and it does not suffer from the problem of units of measurement like variance.
- Although being considered as the best measure of dispersion, standard deviation is hardly applicable in comparing different data sets.

- For a set of observations $x_1, x_2, \ldots, x_n$, the following is the formula for the Standard Deviation,

$$s_1 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right)$$

Properties of Standard Deviation

- It describes the square root of the mean of the squares of all deviations in a data set and is also called the root-mean-square deviation.
- The smallest value of the standard deviation is 0 since it cannot be negative.

- When the data values are similar to each other, then the standard deviation will be very low or close to zero.
- But when the data values vary with each other, then the standard variation is high or far from zero.

Example: Measures of Dispersion

Marks obtained by 8 students in a class test: 10, 19, 12, 21, 18, 20, 11, and 19. Find all measures of dispersion.

Range, R = Highest observation - Lowest observation

$= 21 - 10 = 11$

$$\text{IQR} = Q_3 - Q_1$$
$$= 19.5 - 11.5 = 8$$

Example: Measures of Dispersion

Mean deviation (MD):

First, we have to calculate mean,

$$\bar{x} = \frac{(10 + 19 + 12 + 21 + 18 + 20 + 11 + 19)}{8} = 16.25$$

Then, we construct the following table:

x_i $|x_i - \bar{x}|$

10 6.25 19 2.75

12 4.25

$$MD = \frac{1}{n} \sum_{i=1}^{n} |x_i - \bar{x}|$$

21 4.75 18 1.75 20 3.75 11 5.25

19 2.75 Total 31.5

$= \frac{1}{8} \times 31.5$

$= 3.9375$

The average variation of the marks from the mean mark is 3.9375

Example: Measures of Dispersion

Variance and Standard deviation:

We construct the following table:

$x_i x_i - \bar{x}^2$

10 39.0625 21 22.5625

19 7.5625

12 18.0625

Variance, $S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$

Standard deviation = $\sqrt{\text{Variance}}$

$= \sqrt{19.9232}$

$= 4.4635$

18 3.0625 20 14.0625 11

27.5625 19 7.5625 Total

139.4625

$\frac{1}{8-1} \times 139.4625$

$= 19.9232$

16

Example: Measures of Dispersion

Variance and Standard deviation from Computational formula:

We construct the following table:

x_i x^2_i

10 100 19 361

12 144 21 441 Variance, $S^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} x^2_i - n\bar{x}^2\right)$

$= \frac{1}{n}$

$= 19.9286$

18 324 20 400 11 121 19 361 Standard deviation = $\sqrt{\text{Variance}}$

Total 2252 $= \sqrt{19.9286}$

$\frac{1}{8-1}[2252 - 8(16.25)^2]$ $= 4.4641$

The average variation of the marks from the mean mark is 4.4635

Standard Deviation for grouped data

Class Interval	Class mid value	Frequency
$L_1 - U_1$	x_1	f_1

$L_2 - U_2$	x_2	f_2
$L_3 - U_3$	x_3	f_3
. . .	. . .	. . .
$L_i - U_i$	x_i	f_i
. . .	. . .	. . .

- For a frequency distribution with f_i and x_i denoting the class frequency and class mid value of the ith class (i = 1, 2, . . . , k), the standard deviation can be obtained as:

$$S = {}^{1}\qquad {}^{k}$$

$$\frac{f_i(x_i - \bar{x})^2}{n-1} \quad {}_{i=1}$$

Example

Consider the following data -

Monthly income ('000 Tk)	No. of respondents
5-30	7
30-55	10
55-80	6
80-105	4
105-130	3
Total	30

Find the standard deviation and interpret your result.

Example Solution

Monthly No. of Class

income respondents Midpoint

(’000 tk) (f_i) (x_i) $f_i x_i$ $(x_i - \bar{x})$ $f_i (x_i - \bar{x})^2$ 5-30 7 17.5 122.5 -38.33

10284.32 30-55 10 42.5 425 -13.33 1776.89 55-80 6 67.5 405

11.67 817.13 80-105 4 92.5 370 36.67 5378.76 105-130 3 117.5

352.5 61.67 11409.57 Total 30 1675 29666.67

$$\bar{x} = \frac{f_i x_i}{n} = \frac{1675}{30} = 55.83$$

$$SD = \frac{f_i(x_i - \bar{x})^2}{n-1} = \frac{29666.67}{30-1} = 31.98$$

Average variation of the monthly incomes of the respondents from mean income is 31.98 thousand taka.

Relative measures of dispersion

Coefficient of Variation

- The measures of dispersion like standard deviation are not helpful in comparing variability of two different data sets.
- Difference in the units of measurement and difference in

the magnitude of mean values are the main reason for this.

- That is why a relative measure of dispersion which is free of units of measurement is required.
- Coefficient of variation is one such measure.

Coefficient of Variation Cont.

The ratio of the standard deviation to the arithmetic mean, expressed in percent is known as coefficient of variation (CV). In terms of formula,

$$CV = \frac{S}{\bar{x}} \times 100$$

Coefficient of variation (CV) can be interpreted as the variation per unit mean of a data.

1. The data are in different units
2. The data are in the same units but means are far apart

Example 1: Data are in different units

Suppose that we wise to obtain some insight into whether height is more variable than the weight in the same population. For this purpose, we have the following data obtained from 150 children in a community.

Height Weight

Mean 40 inch 10 kg
SD 5 inch 2 kg
CV 12.5% 20%

Since the coefficient of variation for weight is greater than that of the height, we conclude that weight has more variability than height in the population.

Example 2: Means are far apart

Consider that the blood pressures of a group of patients were measured at two levels: systolic and diastolic, both being measured in the same unit. The results were as follows:

	Systolic	Diastolic
Mean	130 mm Hg	60 mm Hg

SD 15 mm Hg 8 mm Hg

CV 11.5% 13.3%

- Considering the standard deviations, systolic pressure is more variable than the diastolic pressure.
- In terms of relative measured by the CV, the diastolic pressure has the greater variability.

Example 3: Coefficient of Variation

Two factory A and B show the following results about the number of workers and the wages paid to them.

	A	B
No. of workers	5000	6000
Average Monthly Wages	2500	2500

Standard Deviation 9 10

Find in which factory has the greater variability in individual wages.

Solution:

$\bar{x}_A \times 100 = 9$

$CV_A = S_A$

$2500 \times 100 = 0.36\%$

$\bar{y}_B \times 100 = 10$

$CV_B = S_B$

$2500 \times 100 = 0.4\%$

Hence, factory B has greater variability in individual wages. 25

Example 4: Coefficient of Variation

The weekly sales of two products A and B were recorded as follow :

Product A 59 75 27 63 27 28 56

Product B 150 200 125 310 330 250 225

Find out which of the two products shows greater

fluctuation in sales.

Solution

Product A Product B

x x^2 y y^2

59 3481 150 22500 75 5625 200 40000 27 729 125 15625

63 3969 310 96100 27 729 330 108900 28 784 250 62500

56 3136 225 50625 x = 335 x^2 = 18453 y = 1590 y^2 = 396250

Here, $\bar{x}$ = 47.86, $\bar{y}$ = 227.14, S_x = 20.09, S_y = 76.48.

Now,

$CV_x = \frac{S_x}{\bar{x}} \times 100 = \frac{20.09}{}$

$47.86 \times 100 = 41.98\%$

$CV_y = \frac{S_y}{\bar{y}} \times 100 = 76.48$

$227.14 \times 100 = 33.67\%$

Product A has more fluctuation in sales than product B.

Test Yourself

From the analysis of monthly wages paid to employees in two service organizations X and Y, the following results were obtained:

	Organization X	Organization Y

Number of wage earners	550	650
Average monthly wages	5000	4500
Variance of the distribution of wages	900	1600

a. Which organization pays a larger amount as monthly wages? b. Which organization has greater variability in individual monthly wages?

Relative measures (Others)

- Coefficient of Range = $\frac{H-L}{H+L} \times 100$
- Coefficient of Mean Deviation = $\frac{\text{Mean deviation}}{\bar{x}} \times 100$
- Coefficient of Quartile Deviation = $\frac{Q_3-Q_1}{Q_3+Q_1} \times 100$

Shape Characteristics

- The study of shape characteristics of a distribution is of crucial importance in comparing a distribution with other distributions
- By shape characteristic of a distribution we refer to the extent of its asymmetry (known as the measures of skewness) and

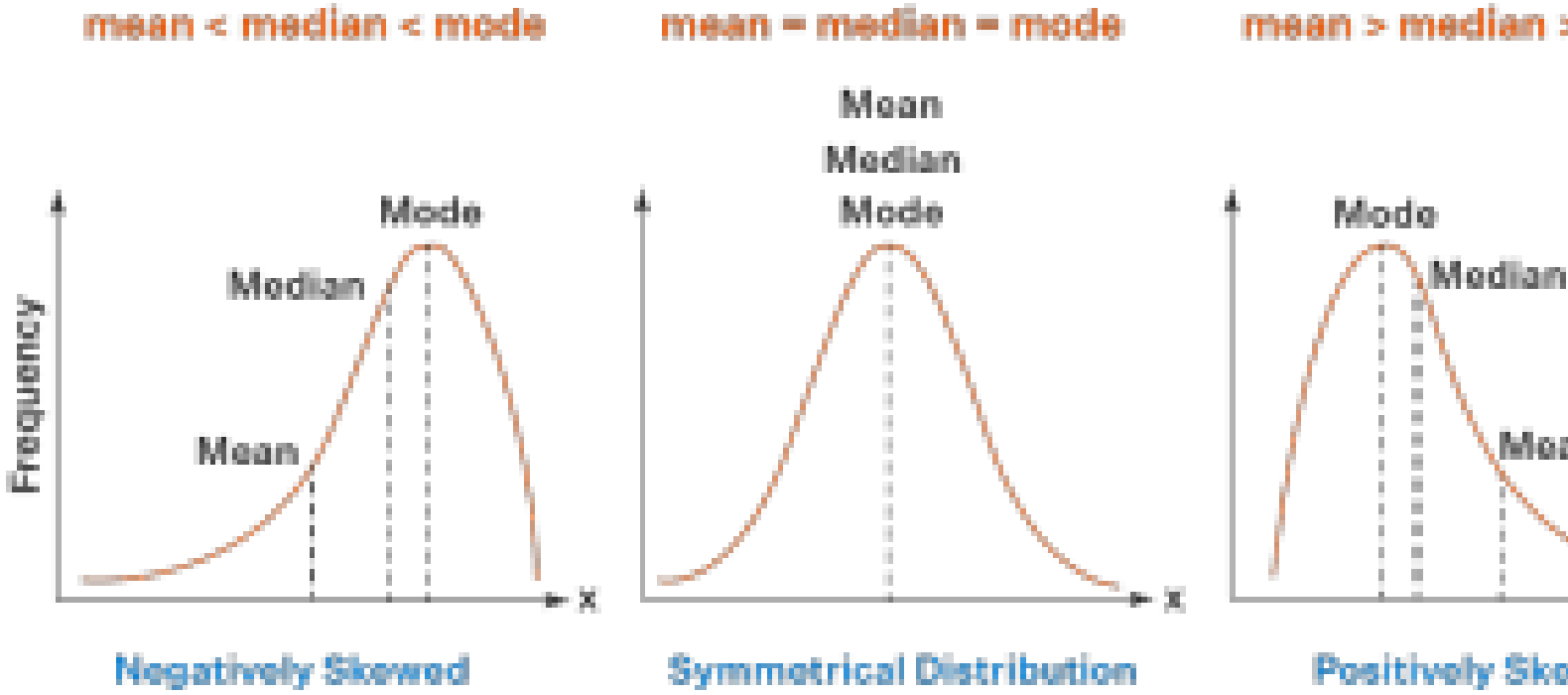
peakedness (known as the measures of kurtosis.)

Skewness

- The term skewness means the lack of symmetry, it may be either positive or negative
- When the skewness is positive, the associated distribution is called positively skewed
- When the skewness is negative, the associated distribution is called negatively skewed

- Simply, for a distribution,
  - Mean > Median > Mode. The distribution is positively skewed
  - Mean < Median < Mode. The distribution is negatively skewed
  - Mean = Median = Mode. The distribution is symmetric.

Skewness of Distributions



Measuring Skewness

Pearson's coefficient for Skewness,

$$SK_p = \frac{Mean - Mode}{SD}$$

$$SD = \frac{3(Mean - Median)}{SD}$$

- $SK_p > 0$; The distribution is positively skewed
- $SK_p < 0$; The distribution is negatively skewed
- $SK_p = 0$; The distribution is symmetric

Measuring Skewness

Bowley's coefficient for Skewness (i.e. quartile skewness coefficient,

$$SK_b = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{Q_3 - Q_1}$$

- $SK_b > 0$; The distribution is positively skewed
- $SK_b < 0$; The distribution is negatively skewed

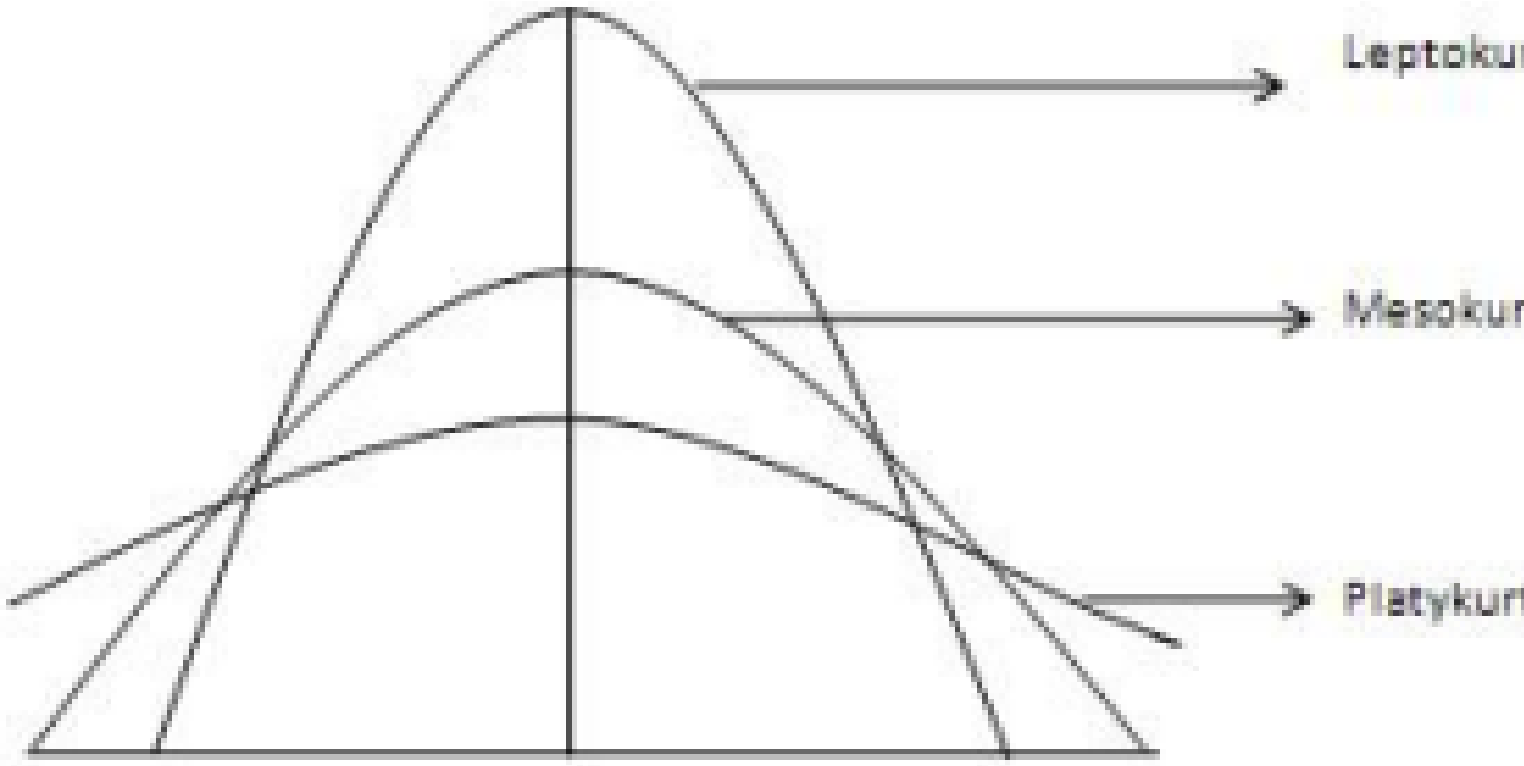
- $SK_b = 0$; The distribution is symmetric

Kurtosis

- Kurtosis as defined by Spiegel, is the degree of peakedness of a distribution, usually taken in relation to a normal distribution • A curve having relatively higher peak than the normal curve, is known as leptokurtic
- A curve, which is neither too peaked nor too flat topped is known as mesokurtic

- A curve that is more flat topped than the normal curve is called platykurtic.

Kurtosis



Summary

- Skewness is a measure of symmetry, or more precisely, the lack of symmetry. A distribution or data set is symmetric if it looks the same to the left and right of the center point. It captures the horizontal spreadness(variability) of the data.

- Kurtosis is a measure of whether the data are heavy tailed or light tailed relative to a normal distribution It helps to determine if the curve is more or less high as compared to the normal curve. It captures the vertical spreadness(variability) of the data.

Test Yourself

1. If for a distribution Mean = 18, Median = 32 and Mode = 36, the distribution is skewed.

2. If for a distribution Mean = 20, Median = 26.4 and SD = 3.3, the distribution is skewed.

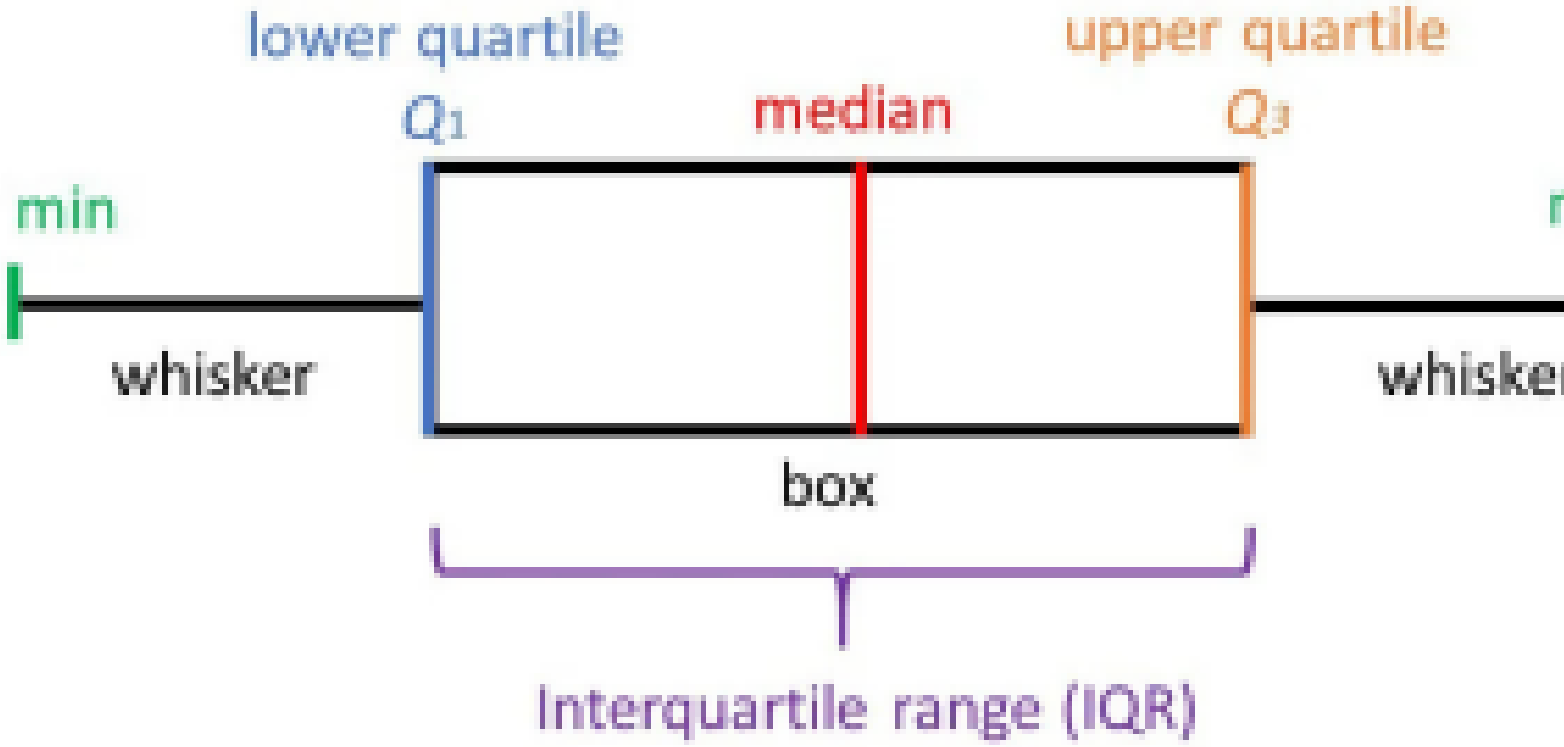
3. If for a distribution, Mean = 35.6, Mode = 24 and SD = 5.2, what is the skewness coefficient of the distribution?

• In descriptive statistics, a box and whisker plot (also called a box plot) is a type of chart often used in explanatory data analysis. • Box plot visually shows the distribution of numerical data and skewness by displaying the data quartiles (or percentiles).

• It shows the five-number summary of a set of data:

1. The minimum score = $Q_1 - 1.5 \times IQR$
2. First quartile (Q_1)
3. Median (Q_2)
4. Third quartile (Q_3)
5. The maximum score = $Q_3 + 1.5 \times IQR$

Box Plot



Box Plot Example

A sample of 10 boxes of raisins has these weights (in grams): 28, 25, 38, 30, 35, 29, 35, 37, 34, 29

Find the five-number summary and make a box plot of the data.

Solution:

Step 1: Order the data from smallest to largest.

Step 2: Find the median. Here, $Q_2 = 32$.

Step 3: Find the quartiles. $Q_1 = 29$ and $Q_3 = 35$.

Step 4: Find IQR = $Q_3 - Q_1 = 35 - 29 = 6$

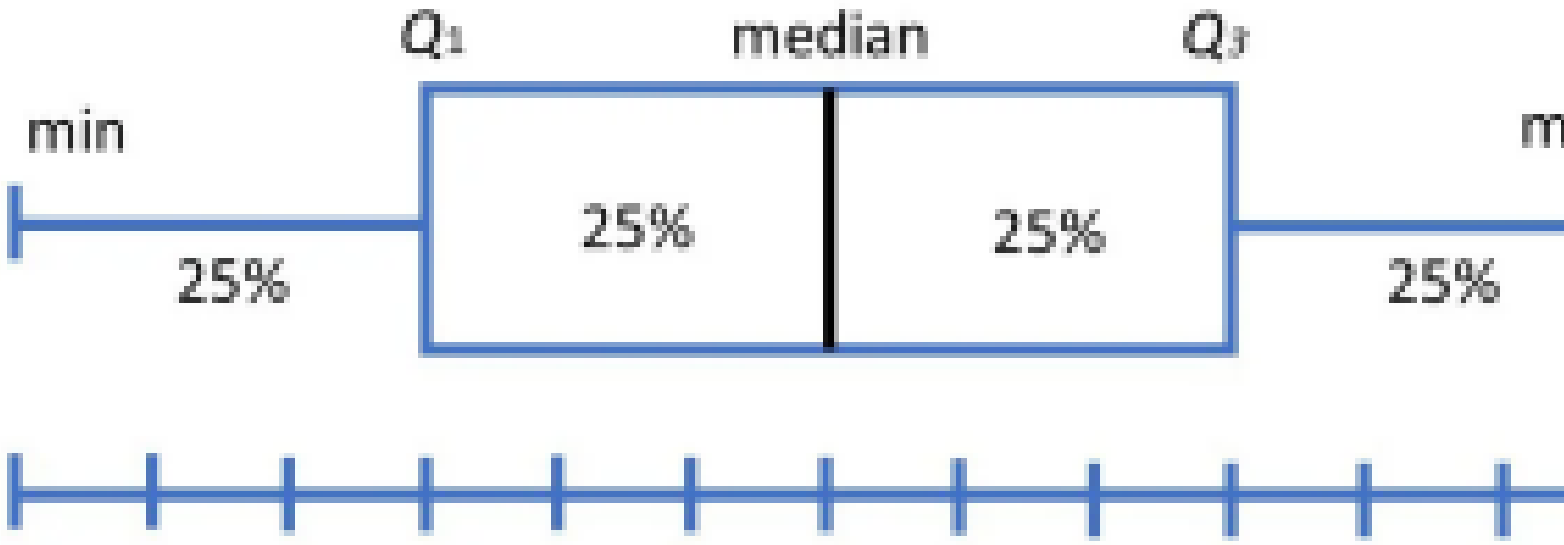
Minimum score = $Q_1 - 1.5 \times IQR = 29 - 1.5 \times 6 = 20$

Maximum score = $Q_3 + 1.5 \times IQR = 35 + 1.5 \times 6 = 44$

Step 5: Construct a box plot. Do it by yourself.

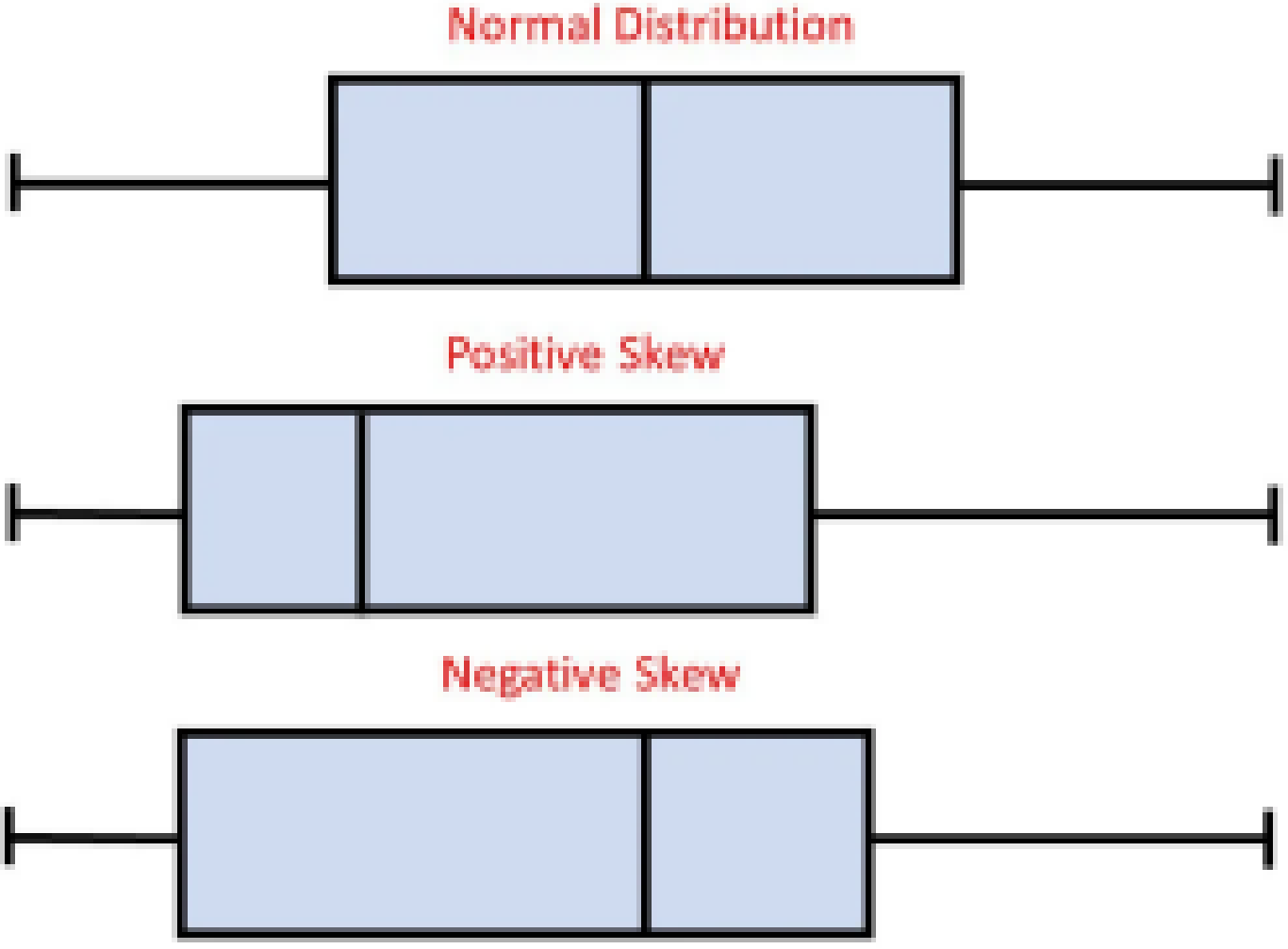
Why Are Box Plots Useful?

1. Box plots divide the data into sections containing approximately 25% of the data in that set.



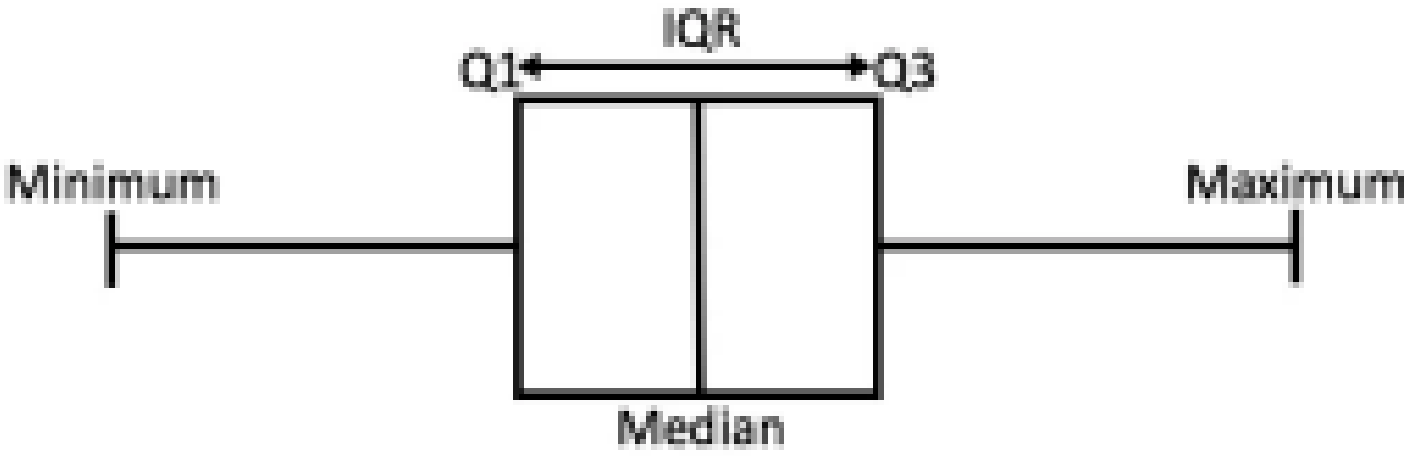
Why Are Box Plots Useful?

2. The box plot shape will show if a statistical data set is normally distributed or skewed.



Why Are Box Plots Useful?

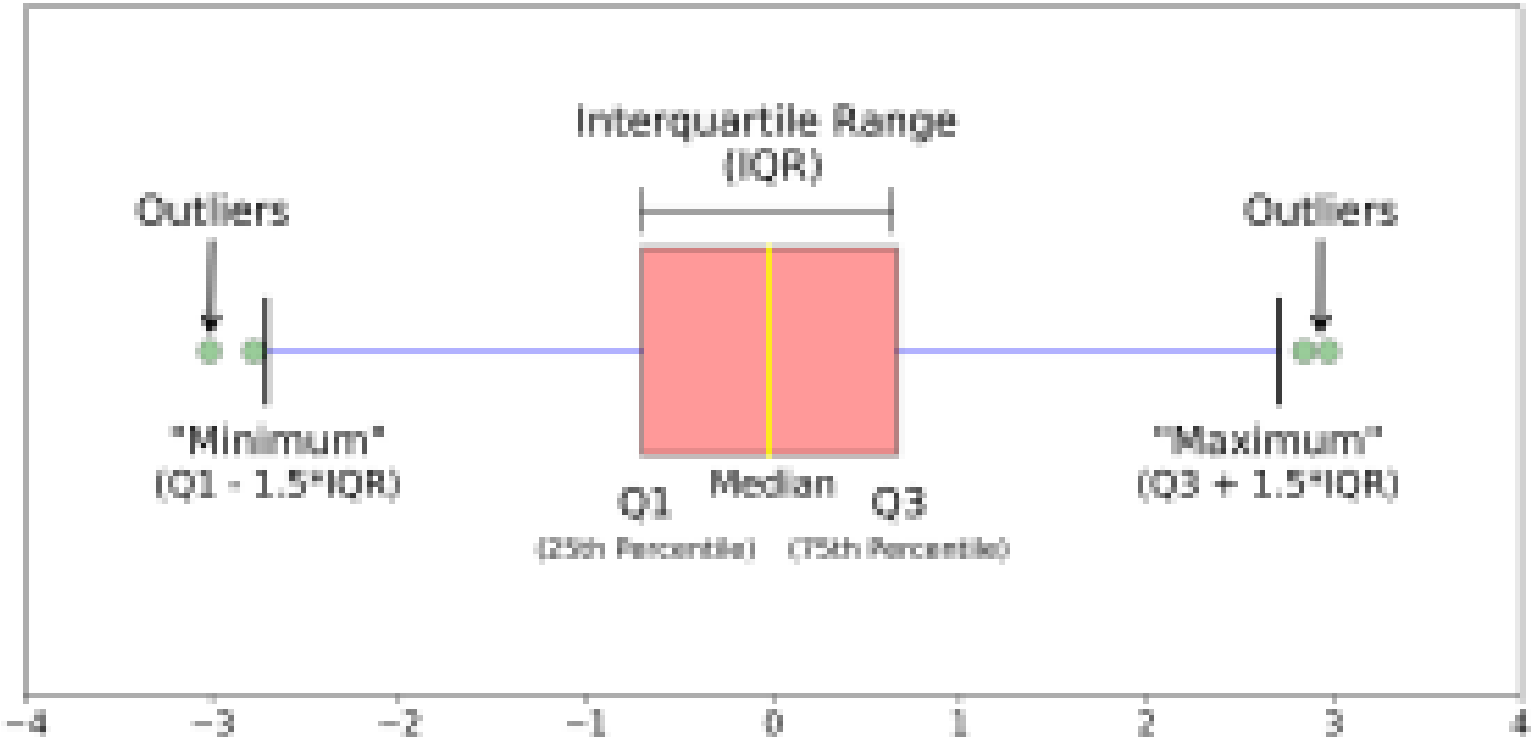
3. Box plots are useful as they show the dispersion of a data set. The interquartile range (IQR) in the box plot showing the middle 50% of data.



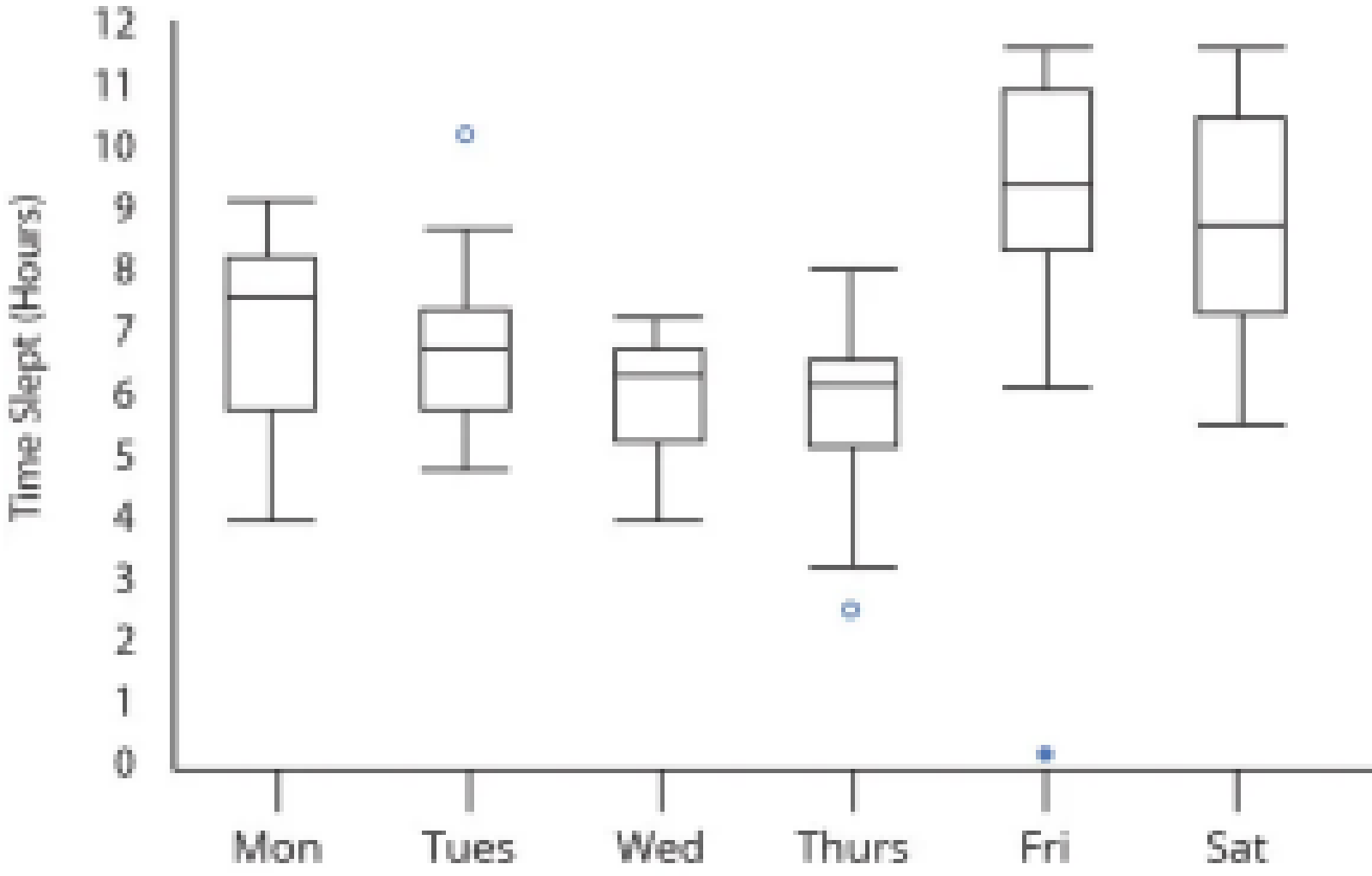
Why Are Box Plots Useful?

4. Box plots are useful as they show outliers within a data set. An outlier is an observation that is numerically distant from the rest of the data.

When reviewing a box plot, an outlier is defined as a data point that is located outside the whiskers of the box plot.



Compare Box Plots



Thank You